

ECx00M-CN&ECx00N-CN

IMS XML Application Note

LTE Standard Module Series

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About the Document

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Contents

About the Document.....	3
Contents	4
Table Index	5
1 Introduction	6
1.1. Applicable Modules	6
1.2. IMS XML Introduction.....	7
1.2.1. IMS XML File Syntax	7
1.2.2. IMS XML Modification Methods.....	8
1.3. IMS XML Update Process.....	11
2 IMS XML AT Commands	13
2.1. AT Command Description	13
2.1.1. Definitions	13
2.1.2. AT Command Syntax.....	13
2.2. Declaration of AT Command Examples	14
2.3. AT+QIMXML Set/Query IMS XML File	14
2.3.1. AT+QIMXML="FORCEUPDATE" Force Update IMS XML File with User XML File.....	14
2.3.2. AT+QIMXML="DEFAULT" Force Update IMS XML File with IMS Default Configuration	15
2.3.3. AT+QIMXML="VERSION" Query IMS XML File Version.....	15
3 Examples	16
3.1. XML File Example	16
3.2. IMS XML Update Methods	17
3.2.1. Update IMS XML through FILE	17
3.2.2. Update IMS XML through FTP	18
3.2.3. Update IMS XML through HTTP	19
4 Appendix References	21

Table Index

Table 1: Applicable Modules.....	6
Table 2: Types of AT Commands	13
Table 3: Related Documents	21
Table 4: Terms and Abbreviations	21

1 Introduction

Quectel ECx00M-CN family and ECx00N-CN family modules support VoLTE whose implementation requires activation of IMS PDN and the parameters specified by the operator at IMS registration. The IMS parameters are stored in IMS XML configuration file. Since the IMS parameters of global operators are different, each operator needs an IMS XML configuration item. Limited by the module storage space, it is impossible to integrate the IMS configurations for all operators. ECx00M-CN family and ECx00N-CN family modules support custom XML configuration files for IMS service interfaces, and you can configure IMS XML files according to actual scenarios to meet the VoLTE requirements of different operators.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
ECx00M	EC200M-CN
	EC600M-CN
	EC800M-CN
ECx00N	EC200N-CN
	EC800N-CN

NOTE

Limited by the flash size of the module, different models of the modules support IMS XML feature differently. For more details, please contact Quectel Technical Support.

1.2. IMS XML Introduction

IMS XML files are specialized configuration files used in IP Multimedia Subsystem (IMS) networks. They contain parameter settings and configurations required for IMS functionality, enabling operators to customize and provision IMS services according to their specific network requirements.

1.2.1. IMS XML File Syntax

1. An example of IMS XML file syntax:

```
<?xml version="1.0" encoding="utf-8"?>
<map version="20200605">
  <oem version="3">
  </oem>
  <user USER_INFO="CUSTOMER">
  </user>
  <operator id="default">
  </operator>
  <operator id="CUCC" id="UNICOM" c="46001" c="46009">
  <string name="KEY_IMS_OPERATOR">cucc</string>
  .....
  </operator>
</map>
```

2. The basic format and characteristics of the elements supported by IMS XML file are described as follows:

- 1) Basic format:

<tag attribute 1="attribute value 1" attribute n="attribute value n">element</tag>. For example:
 <operator id="CUCC" id="UNICOM" c="46001" c="46009">
 <string name="KEY_IMS_OPERATOR">cucc</string>
 </operator>

- 2) Characteristics:

- Attributes and elements can be nested each other. For example, <string name="KEY_IMS_OPERATOR">cucc</string> can be regarded as an element of *operator* and also contains a *name* attribute whose corresponding element is *cucc*.
- The first line of the IMS XML file starts with a fixed format, which indicates the XML version and encoding format. For example, <?xml version="1.0" encoding="utf-8"?> indicates that the XML file version is 1.0 and the encoding format is UTF-8.
- IMS XML file requires a root element. For example,
 <map version="20200605">


```
.....
</map>
```

- The XML file is strictly case-sensitive and the element names cannot be the same, while IMS XML file is case-insensitive and the same element names can be used.

1.2.2. IMS XML Modification Methods

After you receive the IMS XML example file, you need to modify the three attributes according to specific requirements: *oem version*, *user user_info*, and *operator*.

1. *oem version*: IMS XML file version. Range: 1–65535. It is detected at IMS XML update. If it is not configured, an error is reported during IMS XML update. For example, if the IMS XML file version is 3, the configuration is as follows:

```
<oem version="3">
<oem>
```

2. *user user_info*: You can set it to *CUSTOMER*. An error is reported at IMS XML update if the attribute tag is not specified. For example, to set the *user_info* attribute in the XML file, use the following format:

```
<user user_info="CUSTOMER">
</user>
```

3. *operator*: According to the operator's PLMN included in this attribute, you can find the corresponding configuration items required by the IMS registration during the module network registration.

- 1) Attributes of *operator* element:

- *id*: Abbreviation of operator in English.
- *c*: Operator PLMN. This configuration is critical. When used for IMS registration, match the *operator* configuration item corresponding to the IMS XML file according to the registration network PLMN. For example, the XML attribute for China Unicom *operator* is as follows:

```
<operator id="CUCC" id="UNICOM" c="46001" c="46009">
</operator>
```

- 2) The *operator* element contains different configuration items, which are divided into 3 types, each represented as key-value pairs as follows:

- String type:

```
<string name="KEY_DOMAIN_PREFERENCE">PS_ONLY</string>
```
- Boolean type:

```
<boolean name="KEY_AUTH_TYPE_AKA" value="true"/>
```
- Integer type:

```
<integer name="KEY_MTU_SIZE" value="1350"/>
```

3) The main configuration items required for the element contained in *operator* are as follows. If the operator configuration is the same as the default one, the following configuration items can be ignored.

- *KEY_IMS_OPERATOR*: String type. IMS operator name, which is basically the same as *id*. Default value: CMCC.
- *KEY_XCAP_PDP_APN*: String type. APN name used by supplementary service XCAP. Default value: CMNET.
- *KEY_IMS_PDP_TYPE*: String type. IP type of IMS PDP bearer, determined by operator. Default value: IPv4v6.
- *KEY_IMS_PDP_APN*: String type. IMS bearer APN, determined by operator. Default value: IMS.
- *KEY_MTU_SIZE*: String type. MTU size, determined by MTU of 3GPP IMS data package. If the actual data size exceeds the MTU size, TCP is used for transmission, otherwise, UDP is used for transmission. If IMS only supports UDP transmission, set the item to the maximum value. Range: 0–65535. Default value: 1350.
- *KEY_IP_PACKET_SIZE*: Integer type. IP fragment package size of IMS signaling, determined by operators. The package may be discarded by network if the threshold is exceeded. Range: 0–65535. Default value: 1200.
- *KEY_SIP_TIMER_TRING*: Integer type. Maximum waiting time from sending **180 Ringing** to establishing IMS CALL during IMS MO. Default value: 60000. Unit: ms.
- *KEY_SIP_TIMER_TRINGBACK*: Integer type. Maximum waiting time from sending **180 Ringing** to establishing IMS CALL during IMS MT. Default value: 6000. Unit: ms.
- *KEY_SIP_TIMER_RING_INTERVALL*: Integer type. The interval of **RING** URC report during IMS CALL MT. Default value: 5000. Unit: ms.
- *KEY_SMS_CLASS0_STORAGE_ENABLE*: Boolean type. Enable **+CMTI** or **+CMT** report when a message is received in the case of **CLASS0&MT=1**. Default value: TRUE.

An example of operator configuration:

- China Unicom:

```
<operator id="CUCC" id="UNICOM" c="46001" c="46009">
<string name="KEY_IMS_OPERATOR">cucc</string>
<string name="KEY_XCAP_PDP_APN">3gnet</string>
<integer name="KEY_MTU_SIZE" value="1250"/>
<integer name="KEY_IP_PACKET_SIZE" value="1250" />
<boolean name="KEY_XDM_APN_ENABLE" value="true" />
<integer name="KEY_SIP_TIMER_RING_INTERVAL" value="6000" />
<boolean name="KEY_SMS_CLASS0_STORAGE_ENABLE" value="false" />
</operator>
```

NOTE

If the configuration items of operator's VoLTE IMS are not mentioned above, please contact Quectel Technical Support for assistance in adding the custom configuration items and facilitating communication to determine the specific configuration parameters.

1.3. IMS XML Update Process

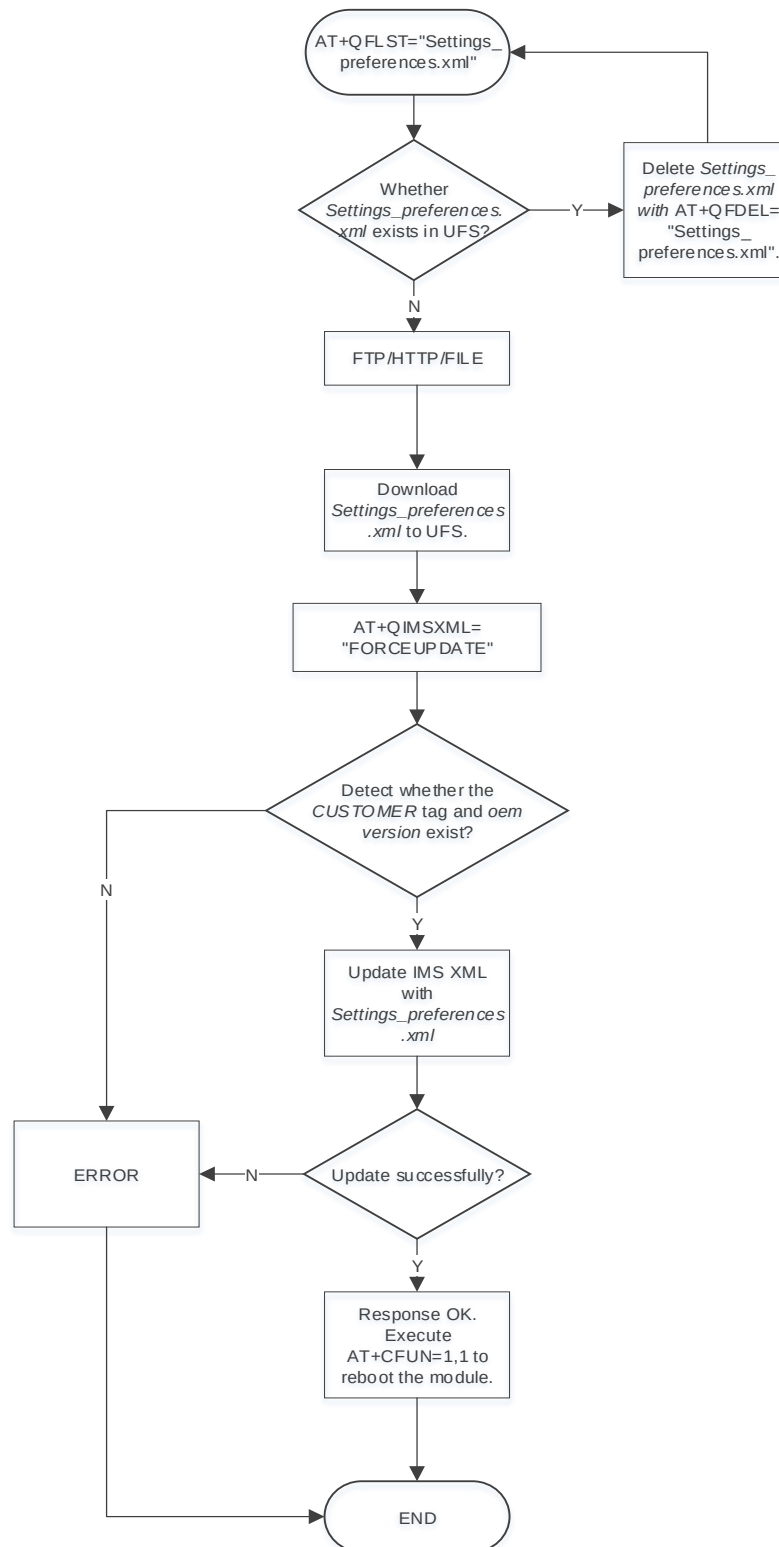


Figure 1: IMS XML Update Process

The modules support uploading the XML file to UFS through FILE, FTP and HTTP and you can execute the related commands to update the XML file to IMS XML storage area, as follows:

Step 1: Upload the XML file to UFS.

- 1) Execute **AT+QFLST="Settings_preferences.xml"** to query whether *Settings_preferences.xml* exists in UFS. If it exists, the file size is returned. And then delete the file with **AT+QFDEL="Settings_preferences.xml"**. If it does not exist, **+CME ERROR: 400** is returned. See *document [1]* for details.
- 2) Upload a new user XML file to UFS if there is no XML file in UFS.
 - Upload an XML file through FILE
Execute **AT+QFUPL=<filename>[,<file_size>[,<timeout>[,<ackmode>]]]** to upload a XML file to UFS. See *document [1]* for details.
 - Download an XML file through FTP
Execute **AT+QFTPGET=<file_name>[,<local_name>[,<startpos>[,<downloadlen>]]]** to download a XML file to UFS through FTP. See *document [2]* for details.
 - Download an XML file through HTTP
Execute **AT+QHHTTPREADFILE=<file_name>[,<wait_time>]** to download an XML file to UFS through HTTP. See *document [3]* for details.

Step 2: Update the XML file.

Force update *Settings_preferences.xml* in UFS to IMS XML storage area with **AT+QIMSEXML="FORCEUPDATE"**.

Step 3: Confirm whether the update is successful.

If **OK** is returned after **AT+QIMSEXML="FORCEUPDATE"** is executed, execute **AT+QIMSEXML="VERSION"** to query the IMS XML file version to confirm whether the update is successful. The updated IMS XML file takes effect after executing **AT+CFUN=1,1** to reboot the module.

NOTE

1. If *Settings_preferences.xml* in the UFS does not contain the *CUSTOMER* configuration or *oem version* is an illegal value, an error is reported, i.e., the XML file update will fail.
2. After **AT+QIMSEXML="DEFAULT"** is executed, the default factory configuration of the IMS XML file is restored, and the user configuration parameters are lost.
3. If the IMS XML file is damaged due to jamming, and the IMS XML file cannot be automatically restored, the module will use the default XML file factory configuration.
4. Disable power-down before **OK** or **ERROR** is returned when **AT+QIMSEXML="FORCEUPDATE"** or **AT+QIMSEXML="DEFAULT"** is executed for the IMS XML update. Otherwise, the IMS XML file will be abnormal and cannot be restored automatically, and the module will use the default factory configuration.

2 IMS XML AT Commands

2.1. AT Command Description

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 2: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

2.3. AT+QIMSEXML Set/Query IMS XML File

AT+QIMSEXML Set/Query IMS XML File	
Test Command AT+QIMSEXML=?	Response +QIMSEXML: "FORCEUPDATE" +QIMSEXML: "DEFAULT" +QIMSEXML: "VERSION" OK
Maximum Response Time	300 ms

2.3.1. AT+QIMSEXML="FORCEUPDATE" Force Update IMS XML File with User XML File

AT+QIMSEXML="FORCEUPDATE" Force Update IMS XML File with User XML File	
Write Command AT+QIMSEXML="FORCEUPDATE"	Response OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

2.3.2. AT+QIMSEXML="DEFAULT" Force Update IMS XML File with IMS Default Configuration

AT+QIMSEXML="DEFAULT" Force Update IMS XML File with IMS Default Configuration	
Write Command AT+QIMSEXML="DEFAULT"	Response OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

2.3.3. AT+QIMSEXML="VERSION" Query IMS XML File Version

AT+QIMSEXML="VERSION" Query IMS XML File Version	
Write Command AT+QIMSEXML="VERSION"	Response +QIMSEXML: "VERSION",<num> OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<num>	Integer type. IMS XML file version. Range: 1–65535.
--------------------	---

3 Examples

3.1. XML File Example

The VoLTE configurations for operators including China Mobile, China Telecom, China Unicom and Italy TIM are as follows:

```
<?xml version="1.0" encoding="utf-8"?>
<map version="20220809">
  <oem version="3">
  </oem>
  <user user_info="CUSTOMER">
  </user>
  <operator id="default">
  </operator>
  <operator id="CMCC" c="46000" c="46002" c="46004" c="46007" c="46008">
    <string name="KEY_IMS_OPERATOR">cmcc</string>
    <stringname="KEY_CONF_FACTORY_URI">sip:mmtel@conf-
factory.ims.mnc000.mcc460.3gppnetwork.org</string>
    <integer name="KEY_SIP_TIMER_RING_INTERVAL" value="6000"/>
    <boolean name="KEY_SMS_CLASS0_STORAGE_ENABLE" value="false"/>
  </operator>
  <operator id="CT" id="CTCC" c="46006" c="46011">
    <string name="KEY_IMS_OPERATOR">ctcc</string>
    <string name="KEY_XCAP_PDP_APN">ctnet</string>
    <integer name="KEY_SIP_TIMER_RING_INTERVAL" value="6000"/>
    <boolean name="KEY_SMS_CLASS0_STORAGE_ENABLE" value="false"/>
    <integer name="KEY_IMS_SESSION_REFRESH_TIMER" value="1800"/>
    <integer name="KEY_SIP_TIMER_TRING" value="20000"/>
    <integer name="KEY_SIP_TIMER_TRINGBACK" value="20000"/>
    <integer name="KEY_IMS_MIN_SE_TIMER" value="1800"/>
  </operator>
  <operator id="CUCC" id="UNICOM" c="46001" c="46009">
    <string name="KEY_IMS_OPERATOR">cucc</string>
    <string name="KEY_XCAP_PDP_APN">3gnet</string>
    <integer name="KEY_MTU_SIZE" value="1250"/>
    <integer name="KEY_IP_PACKET_SIZE" value="1250"/>
  </operator>
</map>
```

```
<boolean name="KEY_XDM_APN_ENABLE" value="true"/>
<integer name="KEY_SIP_TIMER_RING_INTERVAL" value="6000"/>
<boolean name="KEY_SMS_CLASS0_STORAGE_ENABLE" value="false"/>
</operator>
<operator id="TIM_IT" c="22201">
<string name="KEY_IMS_OPERATOR">TIM_IT</string>
<string name="KEY_XCAP_PDP_APN">xcap.tim.it</string>
<integer name="KEY_XCAP_PORT" value="8080"/>
<string name="KEY_XCAP_ROOT_URI">mtasxdms</string>
<string name="KEY_IMS_PDP_TYPE">IPv4</string>
<boolean name="KEY_TRY_DEF_PDP_AFTER_XCAP_PDP_FAIL" value="true"/>
</operator>
</map>
```

3.2. IMS XML Update Methods

3.2.1. Update IMS XML through FILE

The modules support uploading the XML file to UFS through FILE with related commands to update the XML file to IMS XML storage area. See **Chapter 1.3** for details.

AT+QFLST="Settings_preferences.xml"

+QFLST: "UFS:Settings_preferences.xml",1411 //List information of the file named *Settings_preferences.xml* in UFS. File size: 1411 bytes.

OK

AT+QFDEL="UFS:Settings_preferences.xml"

//Delete *Settings_preferences.xml* from UFS.

OK

AT+QFUPL="Settings_preferences.xml",1676,1

//Upload a new user XML file of 1676 bytes.

CONNECT

+QFUPL: 1676,4d0b

OK

AT+QIMSXML="VERSION"

//Query the version of IMS XML file.

+QIMSXML: "VERSION",5

OK

AT+QIMSXML="FORCEUPDATE"

//Force update IMS XML file with user XML file in UFS.

OK

AT+QIMSXML="VERSION"

//Query the version of IMS XML file after update.

+QIMSXML: "VERSION",6

OK

3.2.2. Update IMS XML through FTP

The modules support uploading the XML file to UFS through FTP with related commands to update the XML file to IMS XML storage area. See **Chapter 1.3** for details.

```

AT+QFLST="Settings_preferences.xml"
+QFLST: "UFS:Settings_preferences.xml",1411 //List information of the file named
                                           Settings_preferences.xml in UFS. File size: 1411
                                           bytes.

OK
AT+QFDEL="UFS:Settings_preferences.xml" //Delete Settings_preferences.xml from UFS.
OK
//The configurations related to FTP
AT+QICSGP=1,1,"UNINET","", "",1 //Set PDP context to 1 and China Unicom APN to
                                   "UNINET".

OK
AT+QIACT=1 //Activate PDP context 1.
OK //Activated successfully.
AT+QIACT? //Query the state of PDP context.
+QIACT: 1,1,1,"10.7.157.1"
OK
AT+QFTPCFG="contextid",1 //Set the PDP context ID to 1. The PDP context ID
                           must be activated first.

OK
AT+QFTPCFG="account","test","test" //Set username and password.
OK
AT+QFTPCFG="filetype",1 //Set file type to ASCII.
OK
AT+QFTPCFG="transmode",1 //Set transfer mode to passive mode.
OK
AT+QFTPCFG="rsptimeout",90 //Set the maximum response time (default value:
                             90 seconds).

OK
AT+QFTPOPEN="220.180.239.212",8309 //Login to FTP server.
OK

+QFTPOPEN: 0,0
//Download the XML file from FTP(S) server.
AT+QFTPGET="evin_test/Settings_preferences.xml","UFS:Settings_preferences.xml"
OK

```

```

+QFTPGET: 0,1676 //Downloading successful. File size: 1676 bytes.
AT+QIMSEXML="VERSION" //Query the version of IMS XML file.
+QIMSEXML: "VERSION",5

OK
AT+QIMSEXML="FORCEUPDATE" //Force update IMS XML file with user XML file
                             in UFS.
OK
AT+QIMSEXML="VERSION" //Query the version of IMS XML file after update.
+QIMSEXML: "VERSION",6

OK

```

3.2.3. Update IMS XML through HTTP

The modules support uploading the XML file to UFS through HTTP with related commands to update the XML file to IMS XML storage area. See **Chapter 1.3** for details.

```

AT+QFLST="Settings_preferences.xml"
+QFLST: "UFS:Settings_preferences.xml",1411 //List information of the file named
                                              Settings_preferences.xml in UFS. File size:
                                              1411 bytes.

OK
AT+QFDEL="UFS:Settings_preferences.xml" //Delete Settings_preferences.xml from UFS.
OK
//The configuration related HTTP
AT+QICSGP=1,1,"UNINET","", "",1 //Set PDP context to 1 and China Unicom APN to
                                  "UNINET".

OK
AT+QIACT=1 //Activate PDP context 1.
OK //Activated successfully.
AT+QIACT?
+QIACT: 1,1,1,"10.7.157.1" //Query the state of PDP context.
OK
AT+QHTTPURL=54,80 //Set the length of URL to be accessed and timeout
                   value.

CONNECT
http://11.31.84.134:8300/test/Settings_preferences.xml //Input URL whose length is 54 bytes. (This
                                                         URL is only an example. Input the correct
                                                         URL in use).

OK
AT+QHTTPGET=80 //Send HTTP GET requests with the

```

	maximum response time of 80 seconds.
OK	
+QHTTPGET: 0,200,1676	//If HTTP response header contains CONTENT-LENGTH information, <content_length> is reported.
AT+QHTTPREADFILE="UFS: Settings_preferences.xml",80	//Read response from HTTP(S) server over file.
OK	
+QHTTPREADFILE: 0	//Store HTTP header and body successfully.
AT+QIMSXML="VERSION"	//Query the version of IMS XML file.
+QIMSXML: "VERSION",5	
OK	
AT+QIMSXML="FORCEUPDATE"	//Force update of IMS XML file with user XML file in UFS.
OK	
AT+QIMSXML="VERSION"	//Query the version of IMS XML file after update.
+QIMSXML: "VERSION",6	
OK	

NOTE

For **AT+QICSGP** and **AT+QIACT**, see *document [4]*.

4 Appendix References

Table 3: Related Documents

Document Name
[1] Quectel_ECx00M&ECx00N&EC800K_Series_FILE_Application_Note
[2] Quectel_ECx00M&ECx00N&EC800K_Series_FTP(S)_Application_Note
[3] Quectel_ECx00M&ECx00N&EC800K_Series_HTTP_Application_Note
[4] Quectel_ECx00M&ECx00N&EC800K_Series_TCP(IP)_Application_Note

Table 4: Terms and Abbreviations

Abbreviation	Description
APN	Access Point Name
CUCC	China Unicom Communications Corporation
FTP	File Transfer Protocol
HTTP	Hypertext Transfer Protocol
ID	Identifier
IMS	IP Multimedia Subsystem
MO	Mobile Originated
MT	Mobile Terminated
MTU	Maximum Transmission Unit
OEM	Original Equipment Manufacturer
PDN	Public Data Network
PDP	Packet Data Protocol

PLMN	Public Land Mobile Network
PS	Packet Switch
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
UFS	UNIX File System
VoLTE	Voice over Long-Term Evolution
XCAP	XML Configuration Access Protocol
XML	Extensible Markup Language
